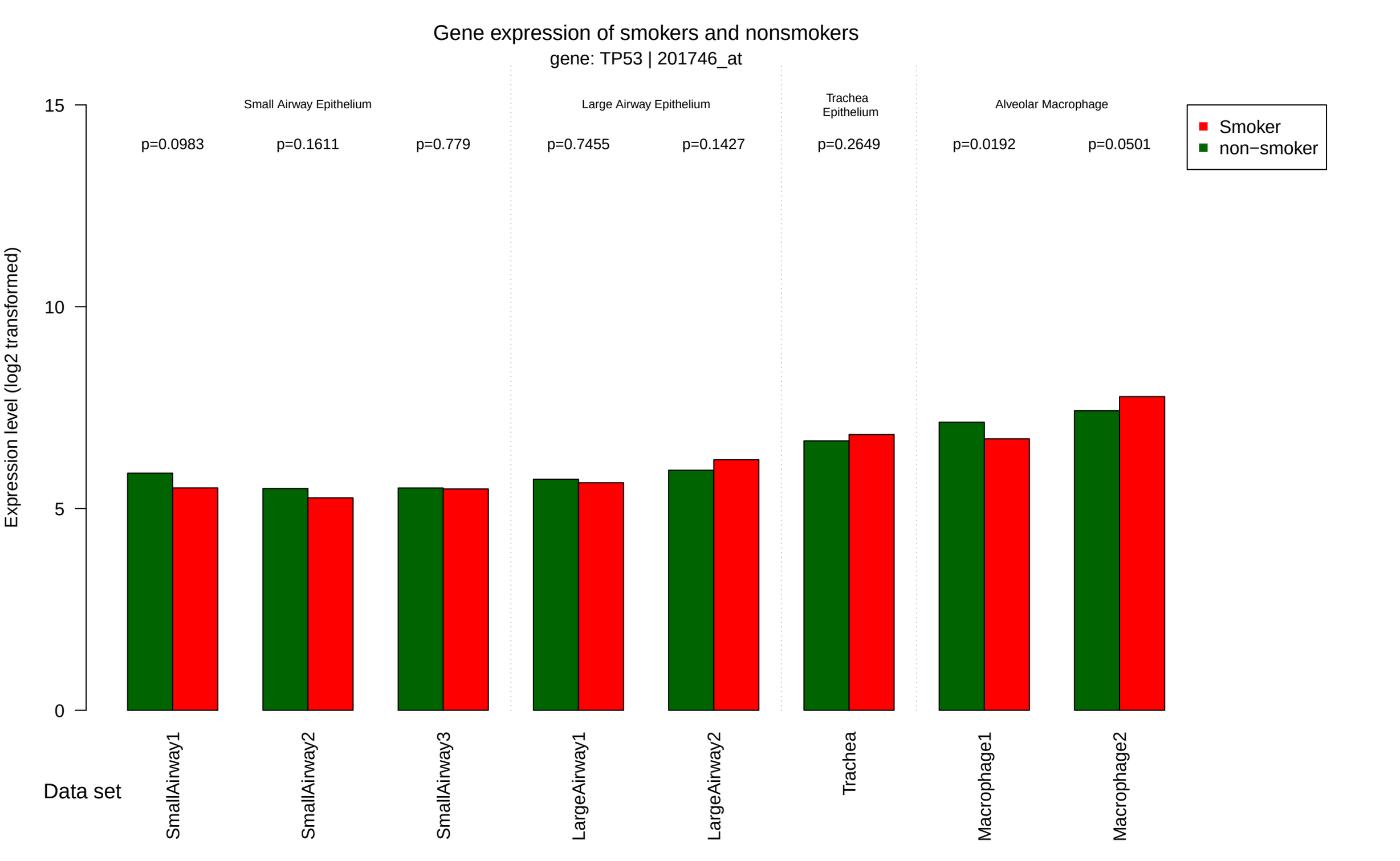
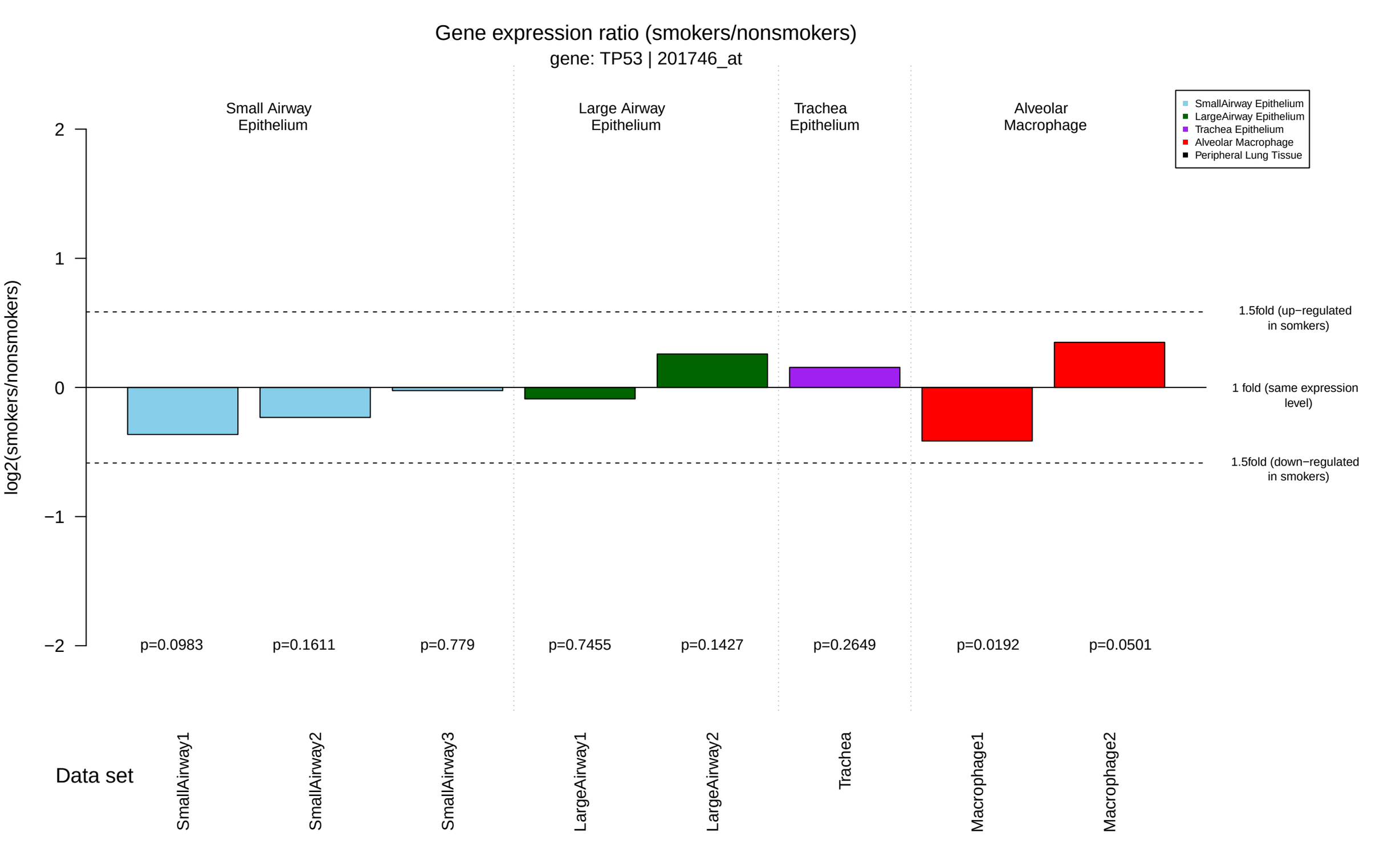
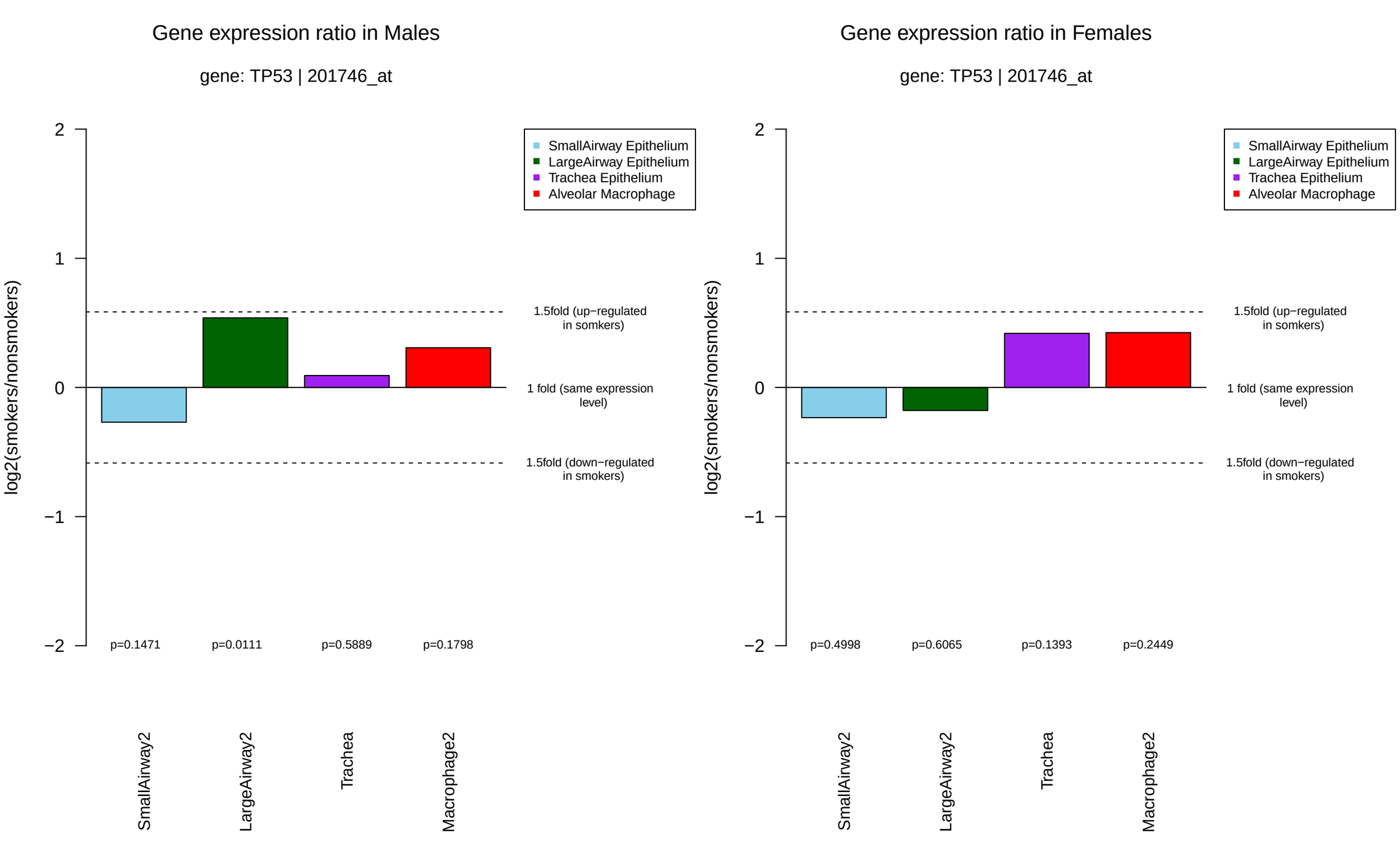
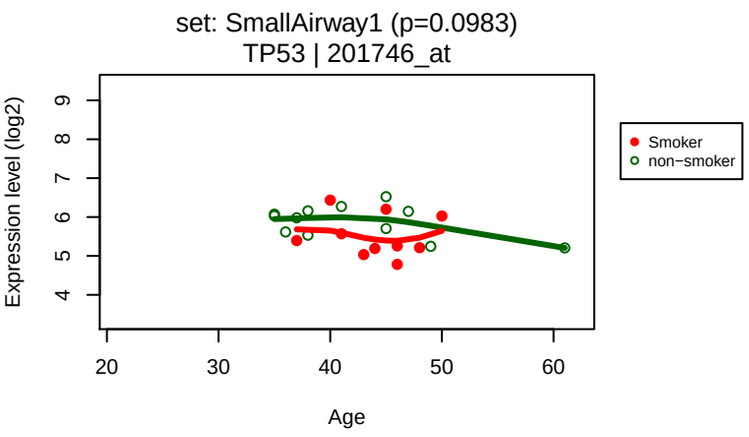
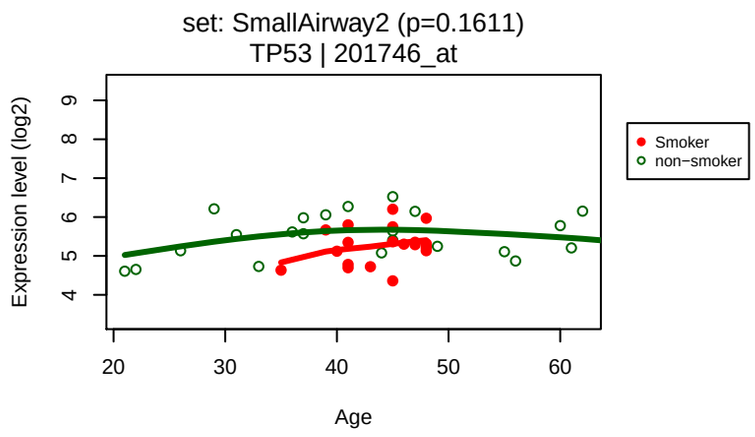
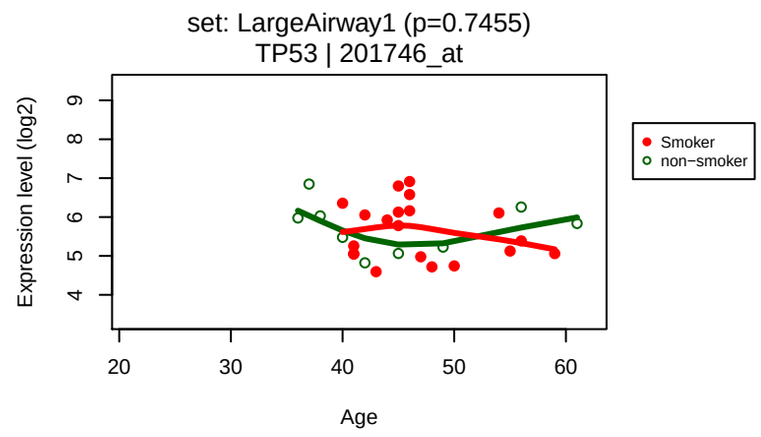

Gene expression of smokers and nonsmokers
gene: TP53 | 201746_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0983
p=0.1611
p=0.779
p=0.7455
p=0.1427
p=0.2649
p=0.0192
p=0.0501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

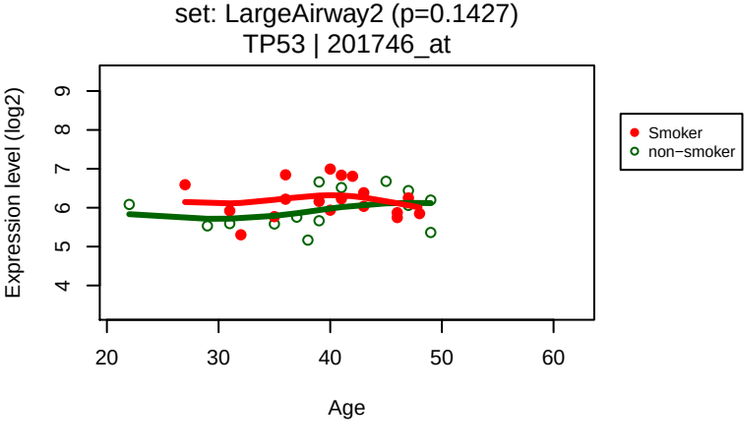

Gene expression ratio (smokers/nonsmokers)
gene: TP53 | 201746_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0983
p=0.1611
p=0.779
p=0.7455
p=0.1427
p=0.2649
p=0.0192
p=0.0501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

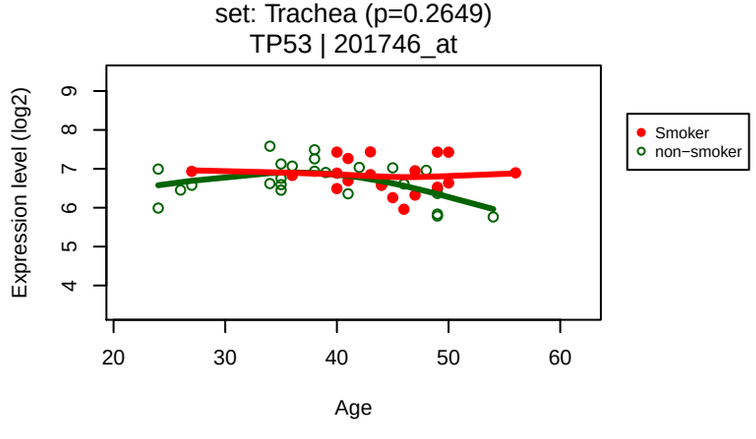

Gene expression ratio in Males
gene: TP53 | 201746_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1471
p=0.0111
p=0.5889
p=0.1798
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TP53 | 201746_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4998
p=0.6065
p=0.1393
p=0.2449
SmallAirway2
LargeAirway2
Trachea
Macrophage2

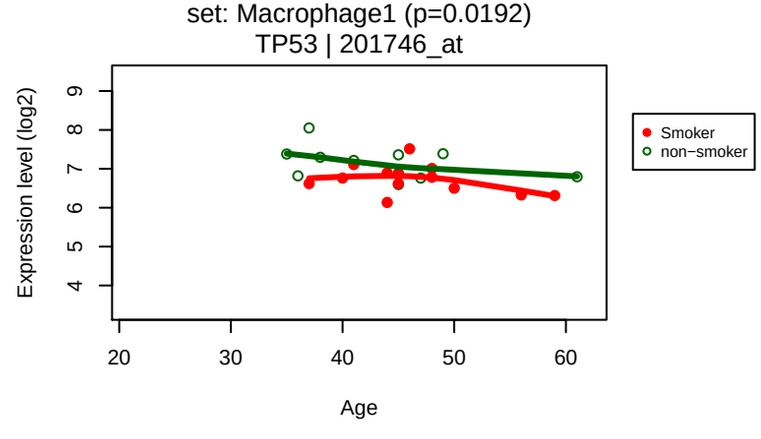

set: SmallAirway1 (p=0.0983)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker

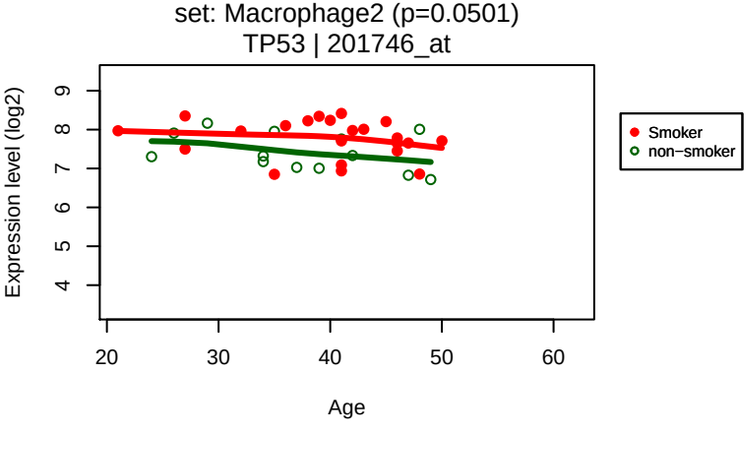

set: SmallAirway2 (p=0.1611)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker

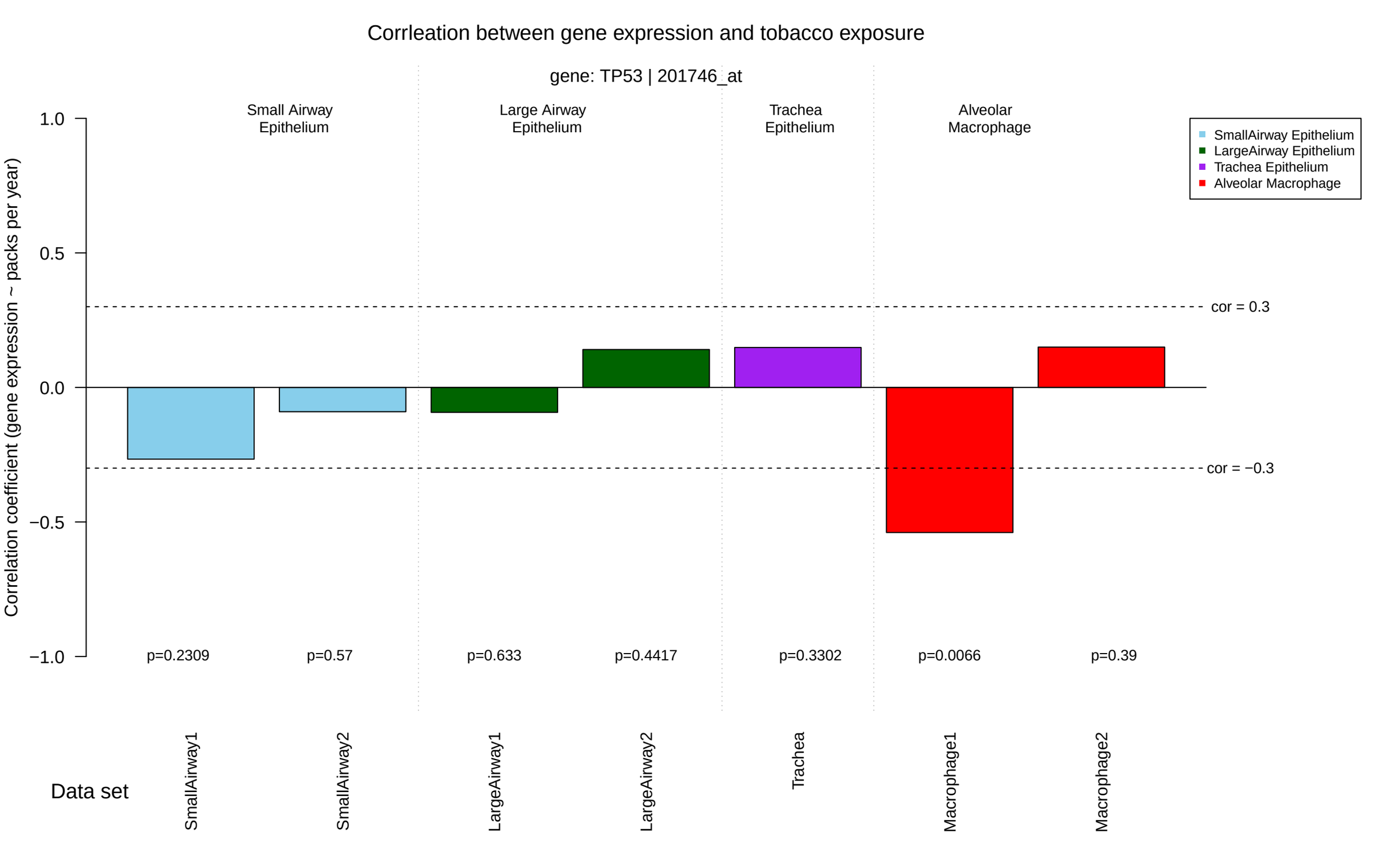

set: LargeAirway1 (p=0.7455)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1427)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2649)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0192)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0501)
TP53 | 201746_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TP53 | 201746_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2309
p=0.57
p=0.633
p=0.4417
p=0.3302
p=0.0066
p=0.39
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2